

PRODUCT DATA SHEET

POLYPROPYLENE

Fibremod™ GB205UC

POLYPROPYLENE COMPOUND, GLASS FIBRE REINFORCED

DESCRIPTION

Fibremod GB205UC is a 20 % chemically coupled glass fibre reinforced polypropylene compound intended for injection moulding. The product is available in standard black 9502.

This material shows good mechanical properties at elevated temperatures.

APPLICATIONS

Fibremod GB205UC has been developed especially for applications in under the bonnet applications

Brackets

Bumper Stiffeners

SPECIAL FEATURES

Long Term Heat Stabilised

PHYSICAL PROPERTIES

Property	Typical Value	Test Method
Density	1040 kg/m ³	ISO 1183
Melt Flow Rate (230°C/2.16kg)	3 g/10min	ISO 1133
Flexural Modulus (2mm/min)	4400 MPa	ISO 178
Tensile Stress at Yield (50mm/min)	75 MPa	ISO 527-2
Charpy Impact Strength, notched (23°C)	11 kJ/m ²	ISO 179/1eA
Charpy Impact Strength, notched (-30°C)	6 kJ/m ²	ISO 179/1eA
Mould Average Shrinkage in flow*	0.35% - 0.45%	Borouge Method
Mould Average Shrinkage cross flow*	1.0% - 1.2%	Borouge Method

*Measured on 150 mm x 90 mm x 3 mm plaques at conditions 23°C. These values are only valid as a reference and not for specification/tool cutting purpose.

*Data should not be used for specification work

PROCESSING CONDITIONS

The actual conditions will depend on the type of equipment used.

This product is easy to process with standard injection moulding machines. To avoid residual humidity from transport or storage, the material should be pre-dried approximately 2h at 80°C.

Following parameters should be used as guidelines:

Holding pressure	30-60 bar
Back pressure	Low to medium

Mould temperature	30 - 50°C
Mass temperature	220 - 260°C
Screw speed	Slow to medium
Flow front speed	100 – 200 mm/s

Specific recommendations for processing conditions can be determined only when the application and type of equipment are known. Please contact your local Borouge representative for such particulars.

STORAGE

Fibremod GB205UC should be stored in dry conditions at temperatures below 50°C and protected from UV-light. Improper storage can initiate degradation which results in odour generation and colour changes and can have negative effects on the physical properties of this product.

More information on storage can be found in Safety Information Sheet (SIS) for this product.

SAFETY

The product is not classified as a hazardous preparation.

Please see our Safety Information Sheet (SIS) for details on various aspects of safety, recovery and disposal of the product, for more information contact your Borouge representative.

RECYCLING

The product is suitable for recycling using modern methods of shredding and cleaning. In-house production waste should be kept clean to facilitate direct recycling.

RELATED DOCUMENTS

The following related documents are available on request, and represent various aspects on the usability, safety, recovery and disposal of the product.

Safety Information Sheet
Statement on chemicals, regulations and standards

STANDARDS

Borouge is certified to various ISO standards, please refer to Borouge.com for more information.

DISCLAIMER

The product(s) mentioned herein are not intended to be used for medical, pharmaceutical or healthcare applications and we do not support their use for such applications.

To the best of our knowledge, the information contained herein is accurate and reliable as of the date of publication, however we do not assume any liability whatsoever for the accuracy and completeness of such information.

Borouge makes no warranties which extend beyond the description contained herein. Nothing herein shall constitute any warranty of merchantability or fitness for a particular purpose.

It is the customer's responsibility to inspect and test our products in order to satisfy itself as to the suitability of the products for the customer's particular purpose.

The customer is responsible for the appropriate, safe and legal use, processing and handling of our products.

No liability can be accepted in respect of the use of Borouge products in conjunction with other materials. The information contained herein relates exclusively to our products when not used in conjunction with any third party materials.

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